Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

GTIIT

Calculus 1 - M
Winter Semester - 2024
Lecture 13 - November 25th
1st Part

Ex. 1. Suppose that f and g are continuous on $[a,b]$ and differentiable on (a,b). Suppose also that $g(a) = f(a)$ and $g'(x) < f'(x)$ for all $a < x < b$.

Prove that $g(b) < f(b)$.

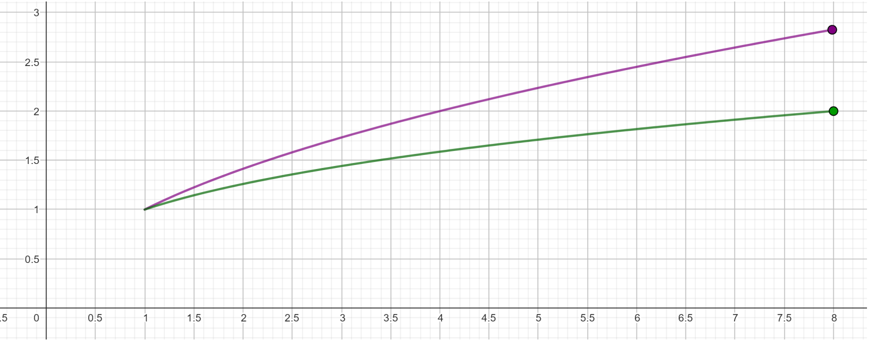
Proof. Apply the previous theorem to $h = f - g$, since:

$$h' = f' - g' > 0 \Rightarrow f - g \text{ is increasing}$$

Hence, $a < b \Rightarrow 0 = (f-g)(a) < (f-g)(b)$

$$g(b) < f(b) \checkmark$$

$0 = h(a)$ and h is increasing $\Rightarrow 0 < h(b)$ ✓



$$h(x) = f(x) - g(x)$$

Ex. 2.

Prove that $\sin(x) < x$ for all $0 < x$.

Proof. Apply the previous result to $f(x) = x$ and
$g(x) = \sin x \;\; on \;\; [0, x]$ for any $0 < x < \pi$:
Take $h(x) = x - \sin x$, then

$$h' = 1 - \cos x > 0 \Rightarrow h(x) = x - \sin x \;\; \text{is increasing}$$

Hence, $0 < x \Rightarrow 0 = h(0) < h(x)$

$$\Rightarrow 0 < x - \sin x \;\; \forall\, 0 < x < \pi$$

Moreover, for any positive $x \in \mathbb{R}$ outside this interval
$\Rightarrow x \geq \pi > 1$

$$\Rightarrow \boxed{\sin x \leq 1 < x \Rightarrow \sin(x) < x \;\; \text{for all} \;\; 0 < x \;\; \checkmark}$$

$\sin(x) < x$ for all $0 < x$ ✓

Ex. 3.

Prove that $|\sin(a)-\sin(b)| \leq |a-b|$ for all a, b.

Solution. By applying the Mean Value Theorem to $f(x)=\sin x$ on the interval (a,b) for any $a<b\in\mathbb{R}$:

$$\frac{\sin b-\sin a}{(b-a)}=\sin' c=\cos c\leq 1 \Rightarrow \sin b-\sin a\leq b-a=|b-a|$$

- If $\sin b-\sin a\leq 0$ then it is less than any positive number
- If $0<\sin b-\sin a$ then it is equal to its absolute value,

so, in both cases

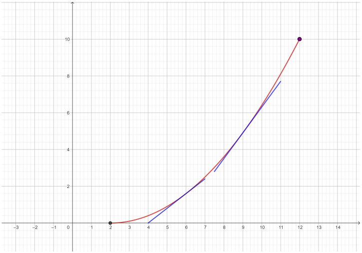
$$\Rightarrow \boxed{|\sin b-\sin a|\leq|b-a| \ \checkmark}$$

What Does f'' Say About f?

Concavity

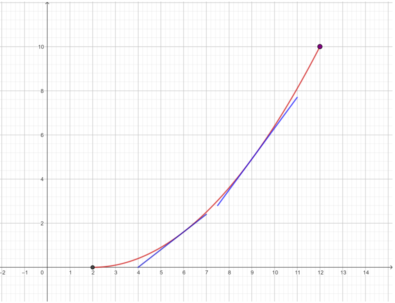
The figures show the graphs of two increasing functions on $[a, b]$, but they look different because they bend in different directions: the concavity is different between them.

$\Rightarrow f''$ allows to distinguish between these two types of behavior



Concavity. Definition

1. If the graph of f lies above all of its tangents on an interval (a, b), then it is called concave upward on (a, b).
2. If the graph of f lies below all of its tangents on (a, b), it is called concave downward on (a, b).



Concave Upward

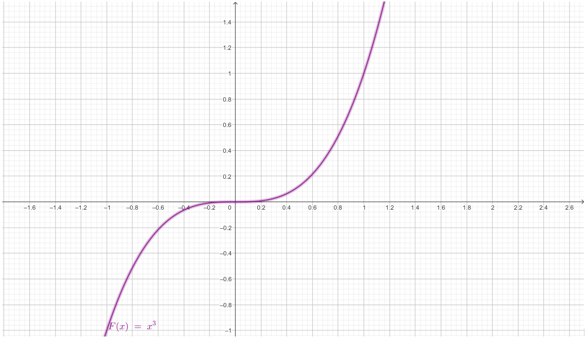
Concave Downward

Concavity and the Second Derivative

Suppose f'', the 2nd derivative of f, exists on (a,b), then

1. If $f''>0$ on (a,b) then f is concave upward on (a,b).
2. If $f''<0$ on (a,b) then f is concave downward on (a,b).

Negative Concavity on $(-\infty, 0)$ Positive Concavity on $(0, +\infty)$



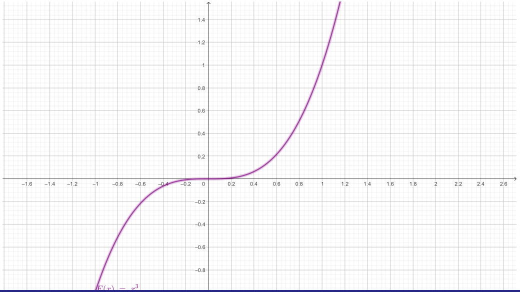
A point $(c, f(c))$ on a curve $y = f(x)$ is called an inflection point if f is continuous at c and:

- the curve changes from concave upward to concave downward at c, or
- the curve changes from concave downward to concave upward at c.

Ex: $f(x) = x^3$ has an inflection point at $x = 0$:

Negative Concavity on $(-\infty, 0)$

Positive Concavity on $(0, +\infty)$



The Second Derivative Test

Suppose f'', the 2nd derivative of f, is continuous on (a, b) and $c \in (a, b)$.

1. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.
2. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.

- The Second Derivative Test is inconclusive when $f''(0) = 0$: at such a point there might be a maximum, there might be a minimum, or there might be neither.

- This test also fails when $f''(0) = 0$ does not exist.

Guide for Complete Analysis and Curve Sketching. For any function $f(x)$, find

1. Dom(f)
2. Symmetries: see if f is odd, even or periodic
3. Intervals of increase and decrease of f.
4. Local maximum and minimum points.
5. Intervals of positive and negative concavity of f.
6. Inflection points.
7. All the asymptotes: vertical, horizontal or slant
8. x- and y- intercepts
9. Use this information to sketch the curve.

Example 4.

Follow the guide to analyze $f(x) = x^4 - 4x^3$ and sketch the curve:

Solution. • f is differentiable on $\mathbb{R}$ because it is a polymonial, hence, 1) and 2) will be analyzed through f':

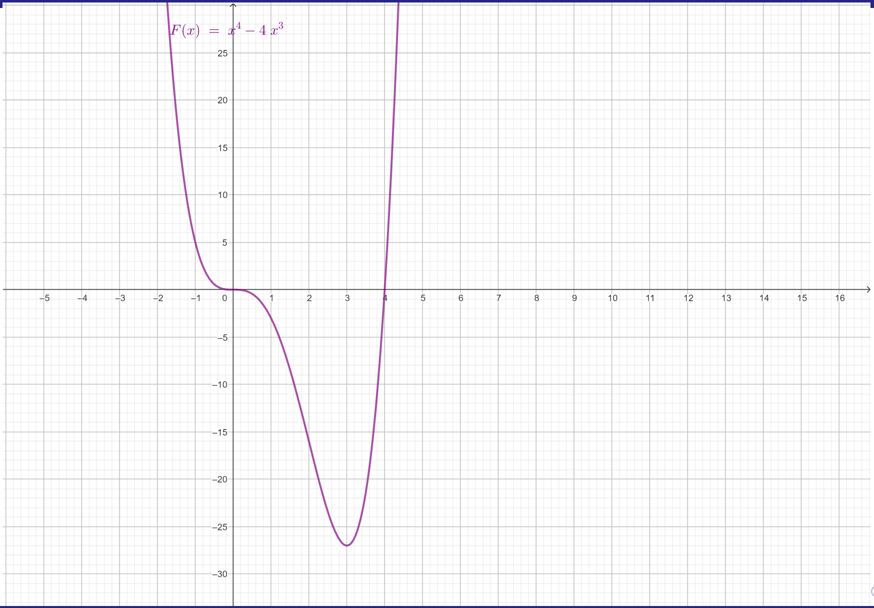
$$\boxed{f'(x)} = 4x^3 - 12x^2 \;\boxed{= 4x^2(x-3)}$$

Then, the critical points are $x = 0$ and $x = 3$

	$(-\infty, 0)$	0	$(0, 3)$	3	$(3, \infty)$
x^2	$+$	0	$+$	$+$	$+$
$(x-3)$	$-$	$-$	$-$	0	$+$
$f'(x)$	$-$	0	$-$	0	$+$

We conclude then

1. f is decreasing on $(-\infty, 3)$
2. f is increasing on $(3, \infty)$.
3. f has a local minimum value at $x = 3$
4. f has no local maximum $\checkmark$


$F(x) = x^4 - 4x^3$

Study of Concavity: use f''

• $f'(x) = 4x^3 - 12x^2$ is differentiable on $\mathbb{R}$ because it is a polymonial, hence, 3) and 4) will be analyzed through f'':

$$\boxed{f''(x)} = 12x^2 - 24x \;\boxed{= 12x(x-2)}$$

Then, the candidates to be inflection points are
$x = 0$ and $x = 2$

Concavity

	$(-\infty, 0)$	0	$(0, 2)$	2	$(2, \infty)$
x	$-$	0	$+$	0	$+$
$(x-2)$	$-$	-2	$+$	0	$+$
$f''(x)$	$+$	0	$-$	0	$+$

We conclude then

1. f has positive concavity on $(-\infty, 0)$ and on $(2, \infty)$
2. f negative concavity on $(0, 2)$.
3. f has inflection points at $x = 0$ and at $x = 2$ ✓

Asymptotes

We analyze the existence of horizontal asymptotes: for this, we need to compute both limits at $\pm\infty$.

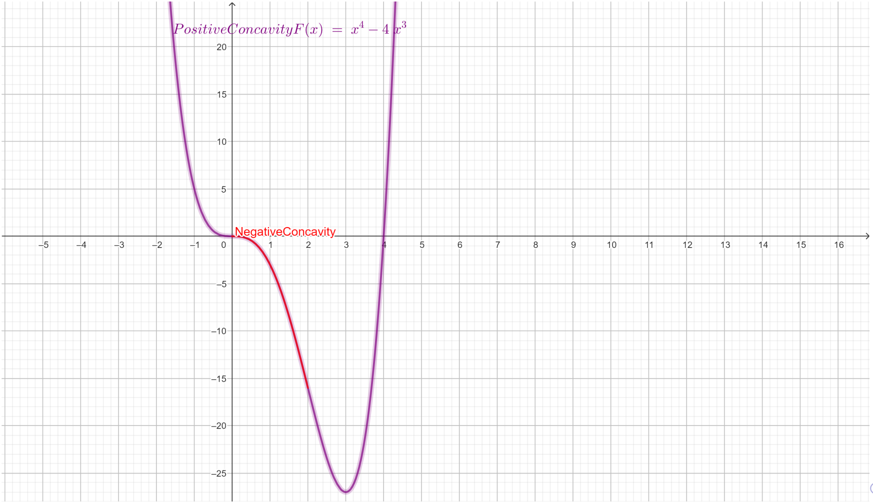
$$\lim_{x\to\pm\infty} f(x) = \lim_{x\to\pm\infty} (x^4 - 4x^3) = \lim_{x\to\pm\infty} \underbrace{x^4}_{\to+\infty} \underbrace{\left(1 - \frac{4}{x}\right)}_{\to 1} = +\infty$$

$\Rightarrow$ There is no horizontal asymptote.

- There is no vertical asymptote, since $Dom(f) = \mathbb{R}$ ✓.

All the collected information

allows to sketch the graph of the curve



L'Hospital's Rule

Suppose f and g are differentiable and $g(x) \neq 0$ on an open interval (a, b) that contains c (except possibly at c). Suppose that

- $\lim_{x\to c} f(x) = 0$ and $\lim_{x\to c} g(x) = 0$, or that
- $\lim_{x\to c} f(x) = \infty$ and $\lim_{x\to c} f(x) = \infty$.

This means that f an indeterminate form of type $0/0$ or ∞/∞. Then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists.

Attention: the right is not the derivative of the quotient

L'Hospital's Rule for Limits at $\pm\infty$

Suppose f and g are differentiable and $g(x) \neq 0$ on an open interval (a, ∞). Suppose that

- $\lim_{x\to+\infty} f(x) = 0$ and $\lim_{x\to+\infty} g(x) = 0$, or that
- $\lim_{x\to+\infty} f(x) = \infty$ and $\lim_{x\to+\infty} f(x) = \infty$.

This means that f an indeterminate form of type $0/0$ or ∞/∞. Then

$$\lim_{x\to+\infty} \frac{f(x)}{g(x)} = \lim_{x\to+\infty} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists or it is $\pm\infty$.

The theorem is also true for for side limits and for f, g differentiable on $(-\infty, b)$ and the limit at $-\infty$

Ex. 6. Check the following limits

a) $\lim_{x\to 1} \dfrac{(x-1)^{4/3}}{\ln(x)} = 0$

b) $\lim_{x\to +\infty} \dfrac{e^{x+2}}{(x+2)^{9/5}} = +\infty$

c) $\lim_{x\to +\infty} \dfrac{\ln(x^2-1)}{\sqrt{x+1}} = 0$

d) $\lim_{x\to -1^+} \dfrac{\ln(x^2-1)}{\sqrt{x+1}} = -\infty$

Solution. c)

Check that both, numerator and denominator, tend to ∞, in order to apply L'Hospital's Rule:

$$\boxed{\lim_{x\to+\infty} \frac{\ln(x^2-1)}{\sqrt{x+1}}} = \boxed{\lim_{x\to+\infty} \frac{\overbrace{\ln(x^2-1)}^{\to+\infty}}{\underbrace{\sqrt{x+1}}_{\to+\infty}}} = \lim_{x\to+\infty} \frac{\frac{2x}{(x^2-1)}}{\frac{1}{2\sqrt{x+1}}}$$

$$= \lim_{x\to+\infty} \frac{4x\sqrt{x+1}}{(x^2-1)}$$

$$= \lim_{x\to+\infty} \frac{4x^{3/2}\sqrt{1+1/x}}{x^2(1-1/x^2)} = \boxed{\lim_{x\to+\infty} \frac{4\sqrt{1+\frac{1}{x}}^{\;\to 1}}{x^{1/2}\underbrace{\left(1-\frac{1}{x}^2\right)}_{\to 1}} = 0 \checkmark}$$

Solution. d)

Be careful: if numerator and denominator do not tend both to 0 or ∞, then it is not possible to apply L'Hospital's Rule:

$$\boxed{\lim_{x\to -1^+} \frac{\ln(x^2-1)}{\sqrt{x+1}}} = \boxed{\lim_{x\to -1^+} \frac{\overbrace{\ln(x^2-1)}^{\to -\infty}}{\underbrace{\sqrt{x+1}}_{\to 0^+}}}$$

$$= \lim_{x\to -1^+} \ln(x^2-1)\cdot\frac{1}{\sqrt{x+1}}$$

$$= \boxed{\lim_{x\to -1^+} \overbrace{\ln(x^2-1)}^{\to -\infty}\cdot\left[\frac{1}{\sqrt{x+1}}\right]^{\to +\infty} = -\infty \ \checkmark}$$

Operations envolving infinite limits

1. Suppose $\lim_{x \to a} f(x) = +\infty$ and $\lim_{x \to a} h(x) = L$, $L > 0$, where $L, a \in \mathbb{R}$ or $a = \pm\infty$, then the following limits exists in the extended sense

$$\lim_{x \to a} \Big(f(x) + g(x) \Big) = +\infty \qquad \lim_{x \to a} \Big(f(x) \cdot g(x) \Big) = +\infty$$

2. Suppose $\lim_{x \to a} f(x) = +\infty = \lim_{x \to a} g(x)$, where $a \in \mathbb{R}$ or $a = \pm\infty$, then the following limits exists in the extended sense

$$\lim_{x \to a} \Big(f(x) + g(x) \Big) = +\infty \qquad \lim_{x \to a} \Big(f(x) \cdot g(x) \Big) = +\infty$$

3. Analogous results follows according with sign rules if $\lim_{x \to a} f(x) = -\infty$ or $\lim_{x \to a} h(x) = L < 0$

Indeterminate Forms

Let $a \in \mathbb{R}$ or $a = \pm\infty$. The following are indeterminations and require algebraic manipulations before taking limits:

1. Suppose $\lim_{x\to a} f(x) = +\infty = \lim_{x\to a} g(x)$, then

   $\lim\limits_{x\to a} \left[f(x) - g(x)\right]$ is an indetermination of kind $\boxed{\infty - \infty}$

2. Suppose $\lim_{x\to a} f(x) = +\infty$ and $\lim_{x\to a} g(x) = 0$, then

   $\lim\limits_{x\to a} \left[f(x)^{g(x)}\right]$ is an indetermination of kind $\boxed{\infty^0}$

3. Suppose $\lim_{x\to a} f(x) = +\infty$ and $\lim_{x\to a} g(x) = 0$, then

   $\lim\limits_{x\to a} \left[f(x) \cdot g(x)\right]$ is an indetermination of kind $\boxed{\infty \cdot 0}$

4. Suppose $\lim_{x\to a} f(x) = 0 = \lim_{x\to a} g(x)$, then

   $\lim\limits_{x\to a} \left[f(x)^{g(x)}\right]$ is an indetermination of kind $\boxed{0^0}$

Indeterminate Forms

Let $a \in \mathbb{R}$ or $a = \pm\infty$. The following are indeterminations and require algebraic manipulations before taking limits:

5. Suppose $\lim_{x\to a} f(x) = +\infty$ and $\lim_{x\to a} g(x) = 0$, then
$$\lim_{x\to a} \left[g(x)^{f(x)}\right] \text{ is an indetermination of kind } \boxed{0^\infty}$$
6. Suppose $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = 1$, then
$$\lim_{x\to a} \left[g(x)^{f(x)}\right] \text{ is an indetermination of kind } \boxed{1^\infty}$$

Attention: We have already solved the case of indetermination of kind 1^∞ applying the limits

$$e = \lim_{x\to 0} \left[1 + x\right]^{1/x} \quad \textit{and} \quad e = \lim_{t\to\infty} \left[1 + \frac{1}{t}\right]^{t}$$

Ex. 7. Rewrite the previous expression as a quotient. Check that both, numerator and denominator, tend to 0 or to ∞, in order to apply L'Hospital's Rule:

$$\lim_{x\to 1^+}\left[\frac{1}{\ln x}-\frac{1}{(x-1)}\right]$$

Solution.

$$f(x)=\frac{1}{\ln x}-\frac{1}{(x-1)}=\frac{x-1-\ln x}{\ln(x)(x-1)}$$

then we can use L'Hospital's Rule since in the new quotient, both, numerator and denominator, tend to 0, so

$$\lim_{x\to 1^+}\left[\frac{1}{\ln x}-\frac{1}{(x-1)}\right]=\lim_{x\to 1^+}\left[\frac{x-1-\ln x}{\underbrace{\ln(x)(x-1)}_{\to 0}}\right]^{\to 0}$$

Next Step: apply L'Hospital's Rule twice

then

$$L = \lim_{x \to 1^+} \left[\frac{1 - \frac{1}{x}}{\underbrace{\frac{1}{x}(x-1) + \ln(x)}_{\to 0}} \right]^{\to 0} = \lim_{x \to 1^+} \left[\frac{1 - \frac{1}{x}}{\underbrace{1 - \frac{1}{x} + \ln(x)}_{\to 0}} \right]^{\to 0}$$

After verifying the hypothesis for the last quotient, we can apply L'Hospital's Rule again:

$$= \lim_{x \to 1^+} \left[\frac{+\frac{1}{x^2}}{\underbrace{+\frac{1}{x^2} + \frac{1}{x}}_{\to 2}} \right]^{\to 1} = \frac{1}{2} \checkmark$$

Indeterminate Forms: Transformations to solve by L'Hospital Rule

Indeterminate form	Conditions	Transformation to $0/0$	Transformation to ∞/∞
$\frac{0}{0}$	$\lim_{x\to c} f(x) = 0,\ \lim_{x\to c} g(x) = 0$	—	$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{1/g(x)}{1/f(x)}$
$\frac{\infty}{\infty}$	$\lim_{x\to c} f(x) = \infty,\ \lim_{x\to c} g(x) = \infty$	$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{1/g(x)}{1/f(x)}$	—
$0 \cdot \infty$	$\lim_{x\to c} f(x) = 0,\ \lim_{x\to c} g(x) = \infty$	$\lim_{x\to c} f(x)g(x) = \lim_{x\to c} \frac{f(x)}{1/g(x)}$	$\lim_{x\to c} f(x)g(x) = \lim_{x\to c} \frac{g(x)}{1/f(x)}$
$\infty - \infty$	$\lim_{x\to c} f(x) = \infty,\ \lim_{x\to c} g(x) = \infty$	$\lim_{x\to c}(f(x) - g(x)) = \lim_{x\to c} \frac{1/g(x) - 1/f(x)}{1/(f(x)g(x))}$	$\lim_{x\to c}(f(x) - g(x)) = \ln \lim_{x\to c} \frac{e^{f(x)}}{e^{g(x)}}$
0^0	$\lim_{x\to c} f(x) = 0^+, \lim_{x\to c} g(x) = 0$	$\lim_{x\to c} f(x)^{g(x)} = \exp \lim_{x\to c} \frac{g(x)}{1/\ln f(x)}$	$\lim_{x\to c} f(x)^{g(x)} = \exp \lim_{x\to c} \frac{\ln f(x)}{1/g(x)}$
1^∞	$\lim_{x\to c} f(x) = 1,\ \lim_{x\to c} g(x) = \infty$	$\lim_{x\to c} f(x)^{g(x)} = \exp \lim_{x\to c} \frac{\ln f(x)}{1/g(x)}$	$\lim_{x\to c} f(x)^{g(x)} = \exp \lim_{x\to c} \frac{g(x)}{1/\ln f(x)}$
∞^0	$\lim_{x\to c} f(x) = \infty,\ \lim_{x\to c} g(x) = 0$	$\lim_{x\to c} f(x)^{g(x)} = \exp \lim_{x\to c} \frac{g(x)}{1/\ln f(x)}$	$\lim_{x\to c} f(x)^{g(x)} = \exp \lim_{x\to c} \frac{\ln f(x)}{1/g(x)}$

Slant Asymptotes

Some curves have asymptotes that are oblique, that is, neither horizontal nor vertical. The line $y = mx + b$ is called a slant asymptote of f at $+\infty$ is

$$\lim_{x\to+\infty}\Big[f(x) - (mx+b)\Big] = 0$$

A similar situation may exist considering the limit at $-\infty$.

The coeffitients are obtained by

- $m = \lim_{x\to+\infty}\left[\frac{f(x)}{x}\right]$
- $b = \lim_{x\to+\infty}\Big[f(x) - mx\Big]$

Example 8. $f(x) = \frac{x^3}{x^2+1}$ has slant asymptotes

Suppose that the line $y = mx + b$ is a slant asymptote of f at $+\infty$, then

$$\boxed{m} = \lim_{x\to+\infty} \frac{f(x)}{x} = \lim_{x\to+\infty} \frac{x^3}{(x^2+1)x} = \lim_{x\to+\infty} \frac{x^3}{x^3} \frac{1}{(1+\frac{1}{x^2})}$$

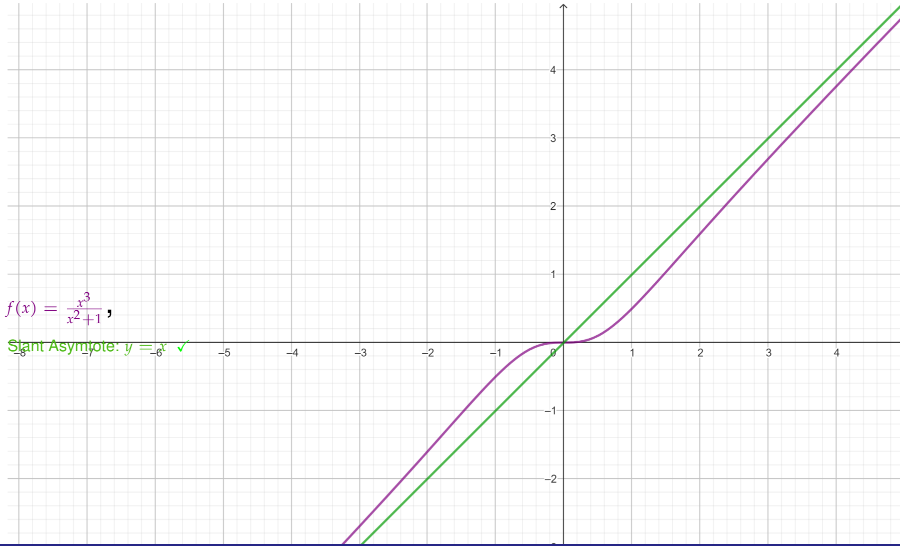
$$= \lim_{x\to+\infty} \frac{1}{(1+\frac{1}{x^2})} \quad \boxed{=1}$$

With this value of m, find

$$\boxed{b = \lim_{x\to+\infty}[f(x) - mx]} = \lim_{x\to+\infty} \left[\frac{x^3}{(x^2+1)} - x\right]$$

$$= \lim_{x\to+\infty} \left[\frac{x^3 - x(x^2+1)}{(x^2+1)}\right] = \lim_{x\to+\infty} \left[\frac{-x}{(x^2+1)}\right] = 0$$

The line $\boxed{y = x}$ is a slant asymptote of f at $+\infty$ and, actually, also at $-\infty$



Optimization Problems.

Steps in Solving Optimization Problems

1. Understand the Problem. Ask yourself: What is the unknown and what are the given quantities?
2. Draw a diagram and identify the given and required quantities on the diagram.
3. Introduce Notation: Assign a variable to the quantity that is to be maximized or minimized

Optimization Problems

Ex 9. A cylindrical can is to be made to hold $1\ liter$ of oil. Find the dimensions that minimize the cost of the metal to manufacture the can.

Solution.

- Let r be the radius and h the height (in centimeters).
- In order to minimize the cost of the metal, we minimize the total surface area of the cylinder:

top, bottom, and sides.

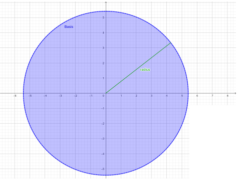
- The sides are made from a rectangular sheet with dimensions $2\pi r$ and h. So, the surface area is

$$\boxed{\text{A = Surface Area } = 2\pi rh + 2\pi r^2}$$

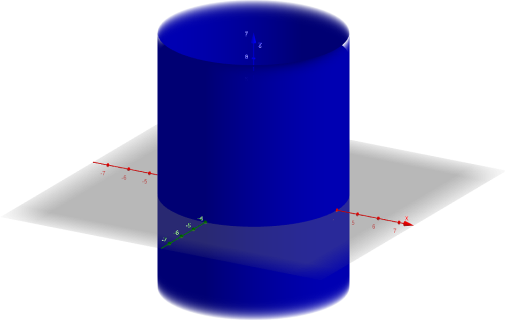
We would like to express A in terms of the variable r:

Minimize the area of a rectangle of a given volume

Bottom = Top with area πr^2



Volume = area of the basis multiplied by the height $= \pi r^2 \cdot h$



- The volume is given as 1 L $= 10^3$ $[cm^3]$. Thus

$$10^3 = V = \pi r^2 h \Rightarrow \boxed{h = \frac{10^3}{\pi r^2}}$$

- Substitution of this into the expression for A gives

$$A(r) = 2\pi r \frac{10^3}{\pi r^2} + 2\pi r^2$$

Then, for $r > 0$, the function to minimize is:

$$\boxed{A(r) = 2000\ \frac{1}{r} + 2\pi r^2}$$

- To find the critical points, we differentiate:

$$\boxed{A'(r) = -2000\ \frac{1}{r^2} + 4\pi r = 0\ \Rightarrow r = \sqrt[3]{\frac{500}{\pi}} \approx 5.42}$$

Search for local extreme values on an open interval

	$(0,\ 5.42)$	$\sqrt[3]{\frac{500}{\pi}}$	$(5.42, \infty)$
$f'(x)$	$-$	0	$+$
$f(x)$	*decreasing*	*absolute minimum*	*increasing*

We conclude then

1. f is decreasing on $(0,\ 5.42)$
2. f is increasing on $(5.42,\ \infty)$.
3. So, f must have a local minimum value at $x = \sqrt[3]{\frac{500}{\pi}} \approx 5.42$, which is, in fact, an absolute minimum value ✓

The area has the lowest value at $x = \sqrt[3]{\frac{500}{\pi}} \approx 5.42$ and this value is $A\left(\sqrt[3]{\frac{500}{\pi}}\right) \approx 553\ [cm^2]$ ✓